

Course Project: Power Conversion Systems – AC/DC, DC/DC Buck-Boost, and DC/DC Flyback Design

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Table of Contents

Project Overview 3
System Operations 3
AC-DC Converter 4
Buck-Boost Converter 5
Potentiometer Calculations 5
Flyback Converter 7
Potentiometer Calculations 7
Simulation 9
Full-Wave Rectifier Simulation 9
Buck-Boost Simulation 10
AC-DC to Buck-Boost Simulation 11
PCB Design and Fabrication 14
Conclusion 16
Appendix 17
Bill Of Materials 17

Project Overview

This project involves the design, simulation, and fabrication of a power conversion system consisting of three major stages: an AC-to-DC rectification front end, a DC-DC buck-boost converter, and an isolated flyback converter. The overall goal of the design is to efficiently convert an AC input source into a regulated, isolated DC output suitable for downstream electronic loads

System Operations

The AC/DC converter begins by taking the ±12 VAC input and passing it through a full-wave diode bridge, producing a pulsating DC waveform that is smoothed by a filter capacitor and regulated down to a stable 3.3 VDC using a linear voltage regulator. This regulated output powers the buck-boost converter, where an analog switching controller drives an inductor in either buck or boost mode depending on duty cycle, with the switching frequency set by external components and the output voltage determined by a feedback network that includes a potentiometer to allow adjustment from 1.2–5.2 V. The flyback converter operates from a rectified 120 VAC input, where a diode bridge produces a high-voltage DC bus that feeds an open-drain switching controller; energy is stored in the transformer during the on-time and transferred to the secondary during the off-time, after which the output is rectified by D2 and filtered to produce the regulated output. As with the buck-boost converter, a potentiometer in the feedback loop adjusts the reference level, allowing the flyback stage to generate an isolated 1.2–5.2 V output.

AC-DC Converter

For the AC-DC converter portion of our project, we used a full-wave bridge rectifier, a type of AC-DC converter that uses four diodes to convert an alternating input voltage into a pulsating DC output. Using a full-wave bridge rectifier is more efficient than using a half-wave rectifier or a single-diode configuration, as it utilizes both halves of the AC cycle. For this project, we designed a basic full-wave bridge rectifier with a linear voltage regulator to set the output voltage to 3.3V.

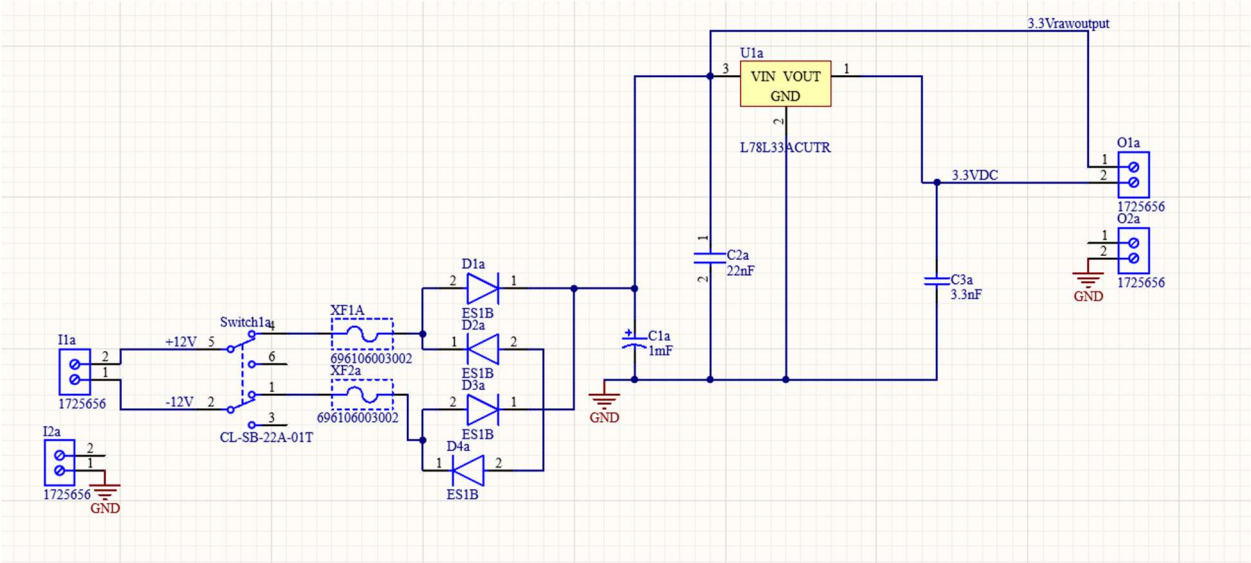


Figure 1. Full-Wave Bridge Rectifier Circuit

Buck-Boost Converter

A Buck-Boost converter is a type of DC-DC converter that can either step down or step up the input voltage to a desired voltage level. Using a Buck-Boost converter is more efficient than using a resistor voltage divider circuit or an amplifier circuit, as it doesn't require an external power supply. For this project, we followed the Altium designer Buck-Boost converter tutorial and modified it to incorporate a potentiometer that allows the output voltage to be adjusted to a desired output voltage.

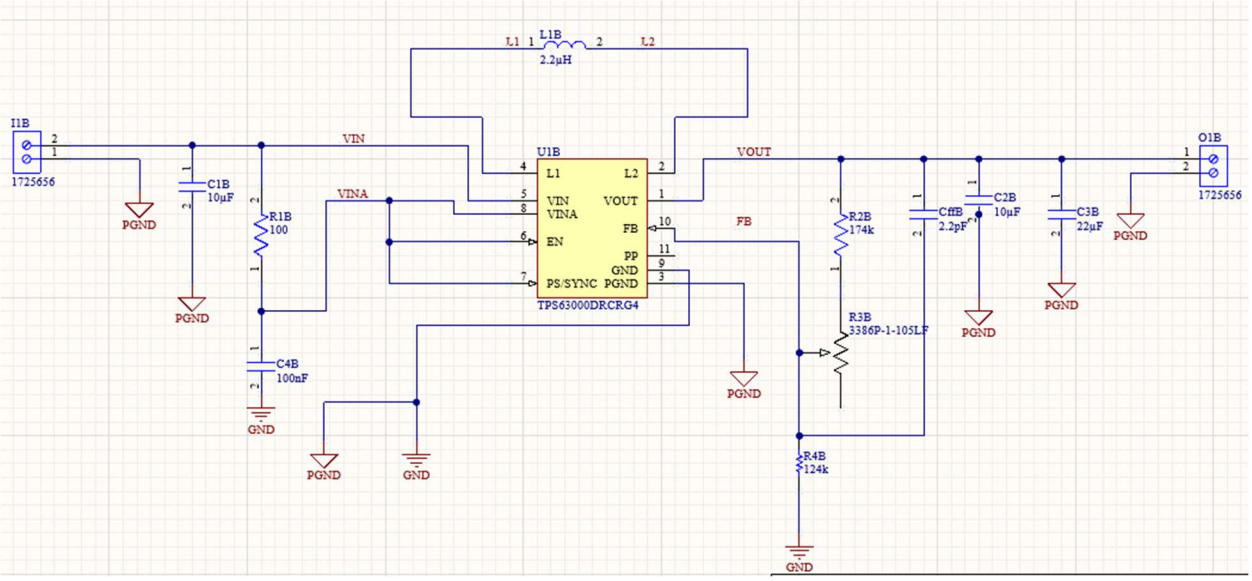


Figure 2. Buck-Boost Schematic

Potentiometer Calculations

The output voltage, $V_{out,}$ can be adjusted by a feedback voltage, V_{FB}, at pin 10 of the TPS63000 chip. The output voltage is given by the formula:

$$V_{out} = \left(\frac{R_1}{R_2} + 1\right) * V_{FB}$$

$$R_1 = \left(\frac{V_{out}}{V_{FB}} - 1\right) * R_2$$

Choose $R_2 = 200k\Omega$, $V_{FB} = 0.5V$ based on datasheet recommendation.

$$R_1 = \left(\frac{5V}{0.5V} - 1\right) * 200k\Omega$$

$$R_1 = 1.8\ M\Omega$$

1.6MΩ is the closest potentiometer value to the calculated value for R_1.

Where R_1 is the top resistor and R_2, is the bottom resistor.

R_1 is replaced by a potentiometer (R3B in Fig. 2). Since a potentiometer's resistance ranges from 0 to the rated resistance, an additional resistor is added to clamp the bottom resistance to prevent V_{FB} from ever equaling V_{out}. The clamping resistor also affects the max resistance; therefore, we choose the next standard potentiometer value of 1MΩ and choose our clamping resistor ($R_{2B,}$ see Fig. 1).

This gives us the formula:

$$V_{out} = \left(\frac{R_{2B} + R_{3B}}{R_{4B}} + 1\right) * V_{FB}$$

Using the formula, the appropriate potentiometer can be found for an output voltage range of 1.2 - 5.2V

$$V_{out} = \left(\frac{R_{2B} + R_{3B}}{R_{4B}} + 1\right) * V_{FB}$$

Finding R2B:

$$R_{2B} = R_{4B}\left(\frac{V_{out}}{V_{FB}} - 1\right) - R_{3B}$$

For V_{out} min, R3B = 0:

$$R_{2B} = R_{4B}\left(\frac{V_{out}}{V_{FB}} - 1\right)$$

$$R_{2B} = 120k\left(\frac{1.2}{0.5} - 1\right) = 168k\Omega$$

Checking for V_{out} max, R_{3B} = 1MΩ:

$$V_{out} = \left(\frac{168k + 1M}{120k} + 1\right) * 0.5$$

$$V_{out} = 4.3V$$

For R_{2B} = 168 kΩ, R_{3B} = 1 MΩ, and R_{4B} = 120 kΩ, our V_{out} ranges from 1.2 – 4.3V, which is within the safety limits of our system (V_{out} min = 1.2V and V_{out} max = 5.2V). Using these calculated resistor values, the closest standard value is picked.

Our final resistor values for the output-voltage feedback circuit are R_{2B} = 174 kΩ, R_{3B} = 1 MΩ, and R_{4B} = 124 kΩ.

Flyback Converter

A flyback converter is a type of isolated switch-mode power supply that efficiently steps down AC voltage to a regulated DC output using transformer-based energy storage and release. Using a flyback converter is more compact and versatile than using linear regulators or buck converters. For this project, we followed the Altium Designer flyback converter tutorial and modified it to include a potentiometer in the feedback loop to regulate the output voltage between 3.3V and 5V.

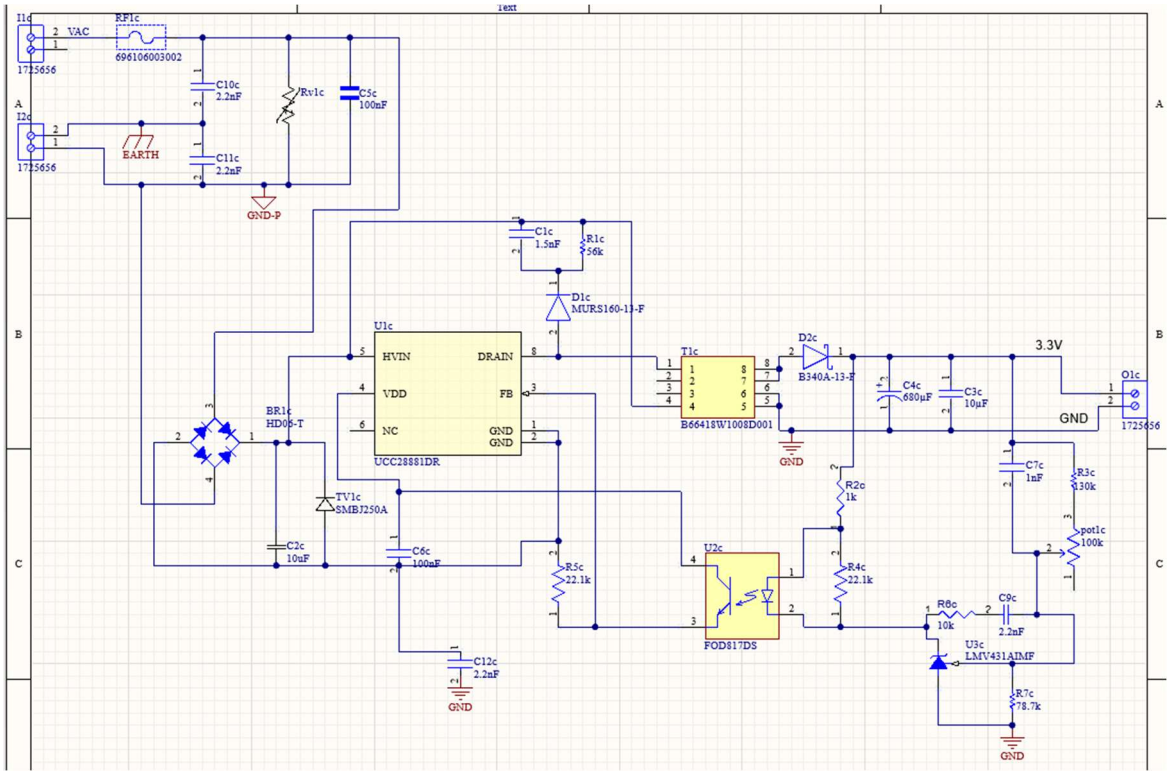


Figure 3. Flyback Converter Schematic

Potentiometer Calculations

$$V_o = V_{ref}\left(1 + \frac{R_3}{R_9}\right)$$

For Vo min (3.3V):

$$V_{o\,min} = 3.3 = 1.24\left(1 + \frac{R_3}{78.7k}\right)$$

$$3.3 = 1.24 + \frac{1.24R_3}{78.7k}$$

$$R_3 = 130.74k\Omega$$

For Vo max (5V):

$$V_{o\,max} = V_{ref}\left(1 + \frac{R_3 + R_{pot}}{R_9}\right)$$

$$V_{o\,max} = 5 = 1.24\left(1 + \frac{130.74k + R_{pot}}{78.7k}\right)$$

$$R_{pot} = 107.895k\Omega$$

Simulation

Simulations in LTspice were carried out to verify the operation of our circuit. To simulate the components of our circuit, we had to create custom LTspice models based on PSPICE models available for our components.

Full-Wave Rectifier Simulation

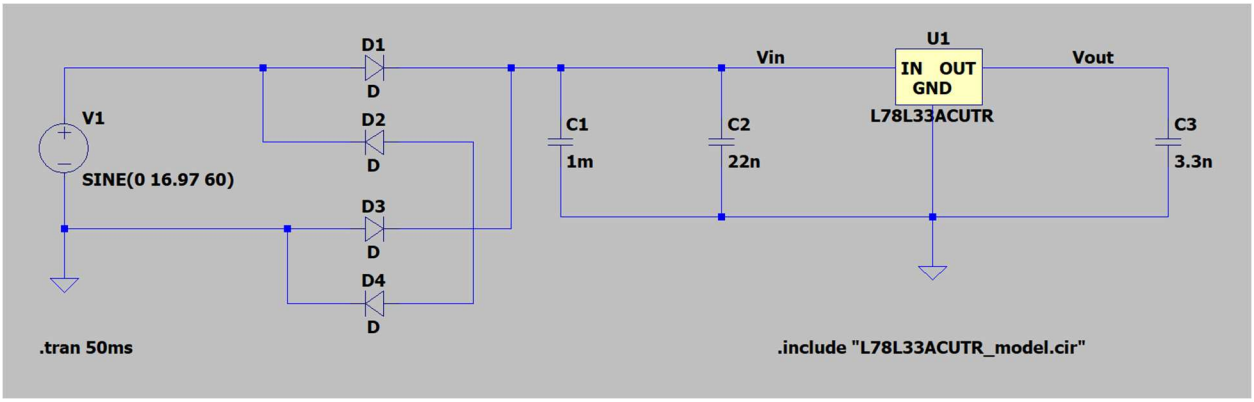


Figure 4. LTspice Simulation of Full-Wave Rectifier Circuit with Linear Voltage Regulator

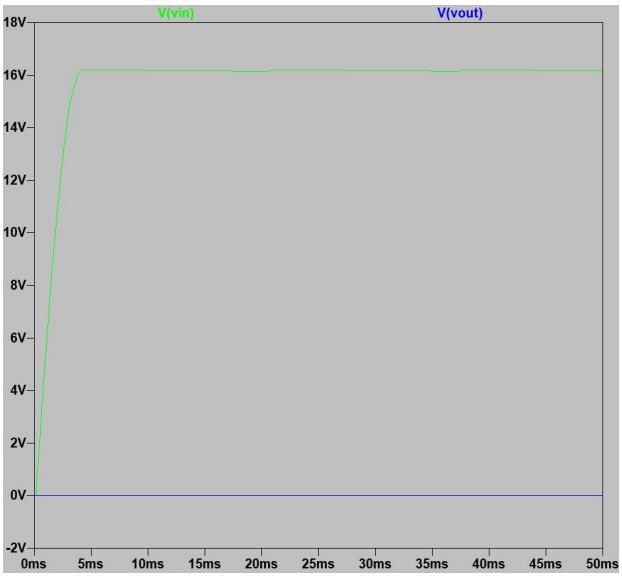


Figure 5. Full-Wave Rectifier Transient Response

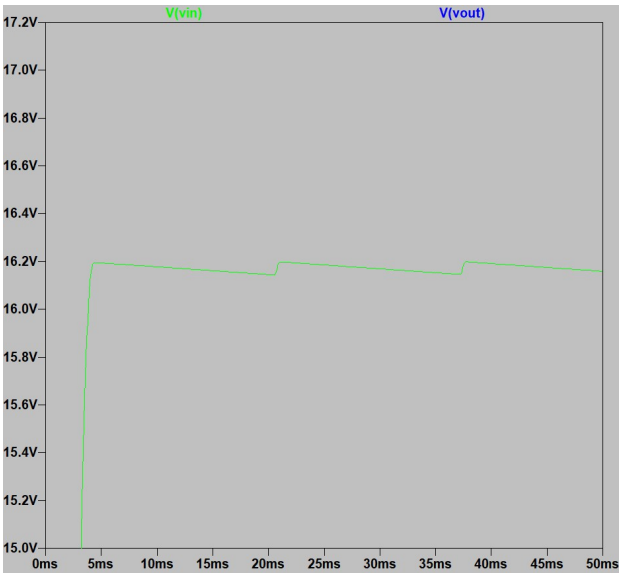


Figure 6. Full-Wave Rectifier Ripple Voltage

Fig. 5 shows the transient response of the full-bridge rectifier circuit with an input sinusoidal wave at 16.97 V_{AC}. The full-bridge rectifier output settles within 5 ms with minimal output voltage ripple. Fig. 6 shows the voltage ripple at the output of the rectifier ($V_{ripple} < 3\%\ V_{out}$).

Buck-Boost Simulation

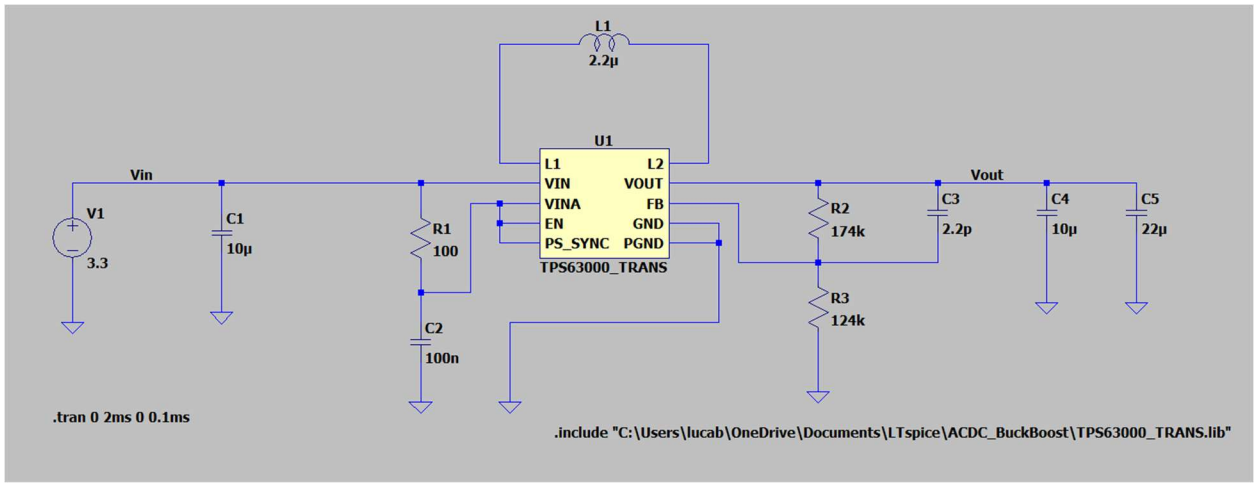


Figure 7. LTspice Simulation of Buck-Boost Converter

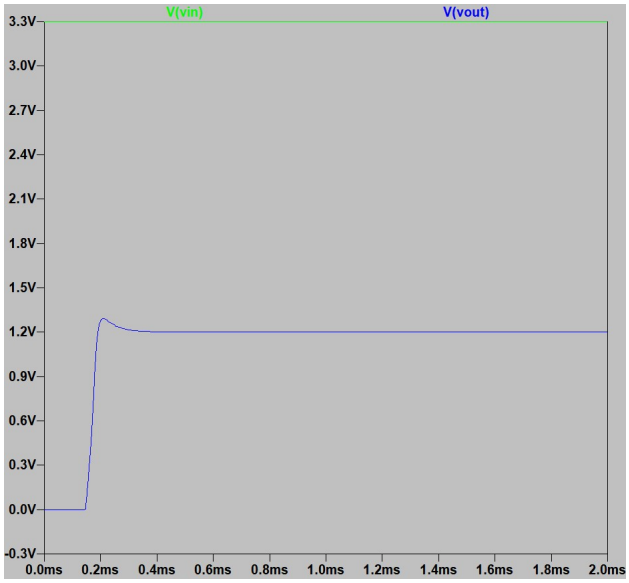


Figure 8. Transient Response (Min Potentiometer Value)

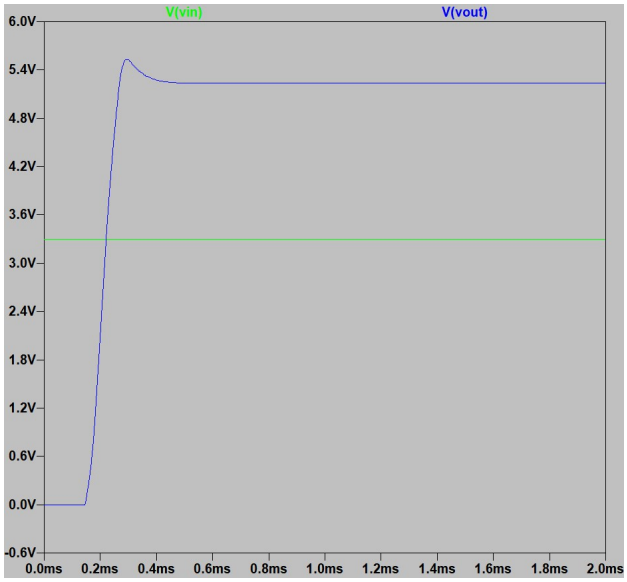


Figure 9. Transient Response (Max Potentiometer Value)

Figures 8 and 9 show the transient response of the Buck-Boost circuit in both the min and max potentiometer value configurations. The response quickly rises with a slight overshoot before settling down to the desired V_{out}. In both configurations, the rise and settling time are short (about 0.1ms), almost negligible. The V_{out} also shows no ripple voltage as expected for this design (V_{ripple} < 1% V_{out}).

AC-DC to Buck-Boost Simulation

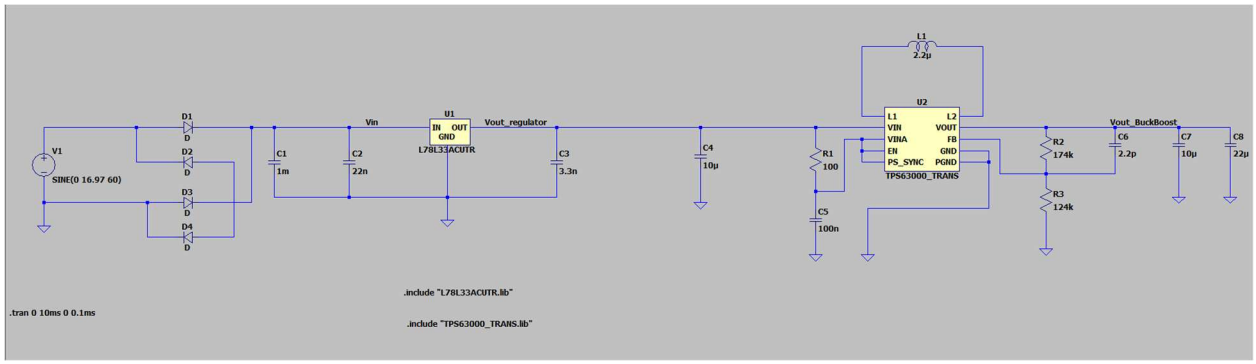


Figure 10. LTspice Simulation of AC-DC to Buck-Boost

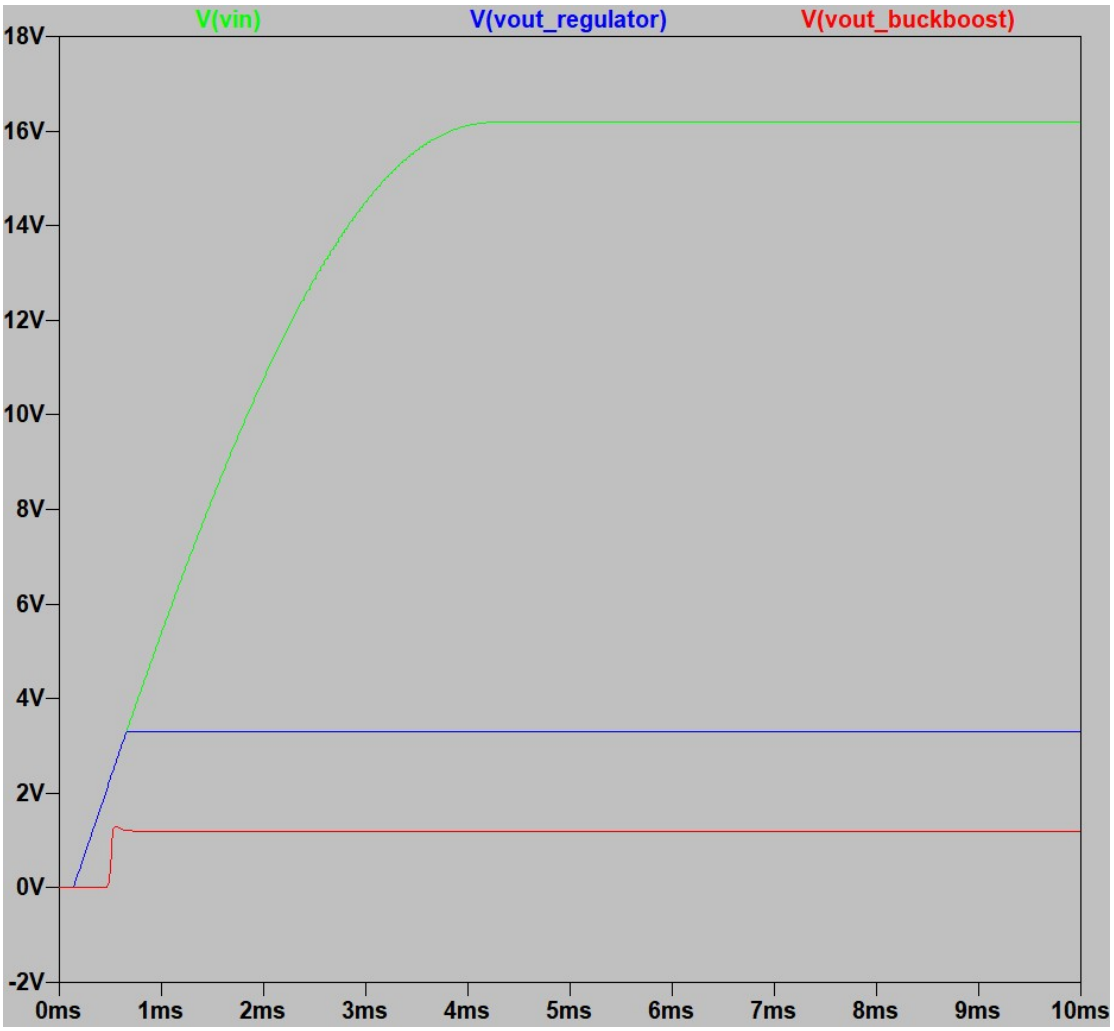


Figure 11. Transient Response (Min Potentiometer Value)

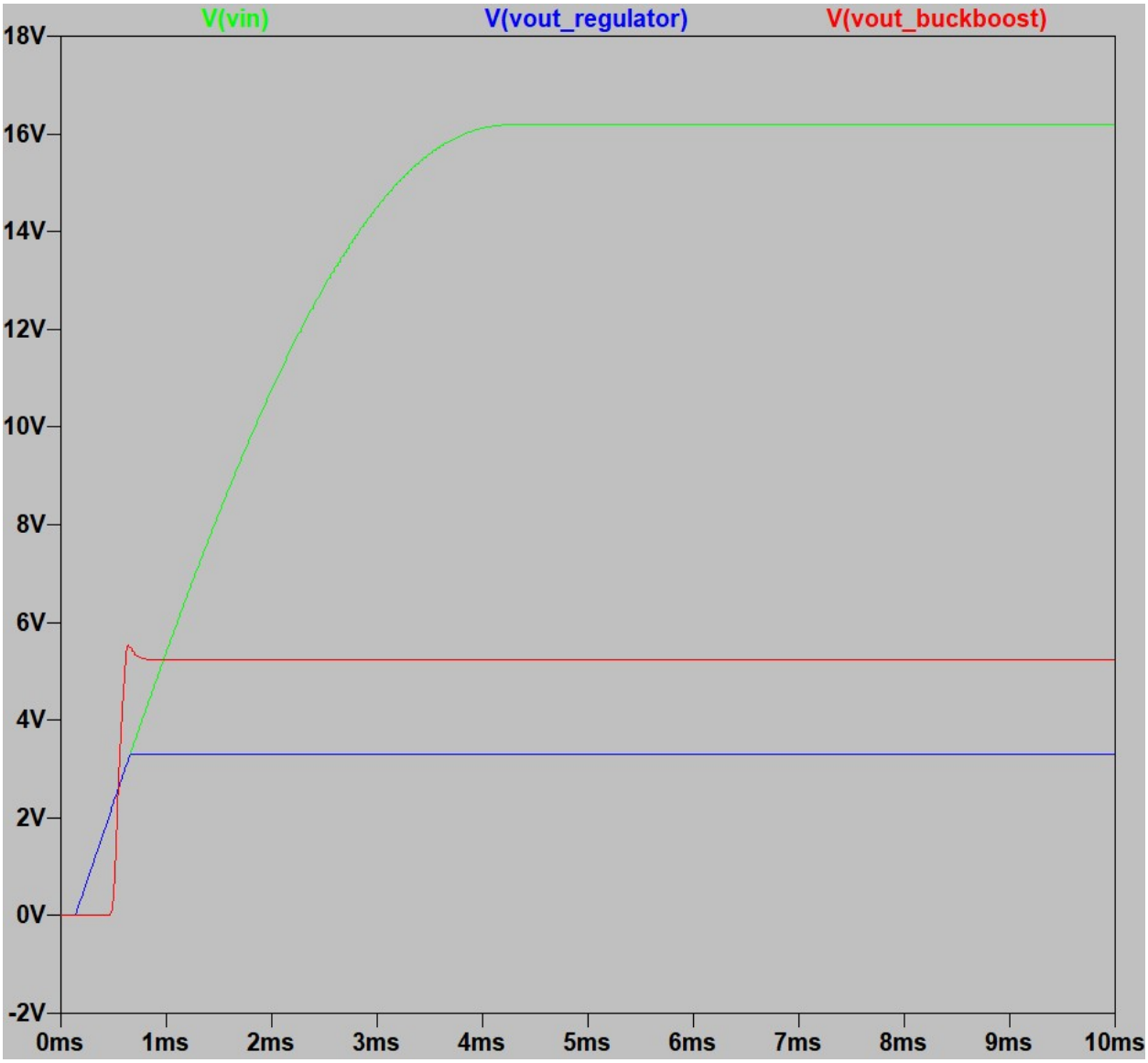


Figure 12. Transient Response (Max Potentiometer Value)

Figures 11 and 12 show the transient response of the full-bridge rectifier and the Buck-Boost together in both the min and max potentiometer configurations, respectively. As seen in Figs. 11 and 12, the rise and settling time of the Buck-Boost output is very short compared to the settling time of the full-bridge rectifier. This is interesting to observe, as the Buck-Boost engages as soon as the output of the linear voltage regulator reaches 2V. This happens considerably faster compared to the time it takes the full-bridge rectifier circuit to arrive at steady-state.

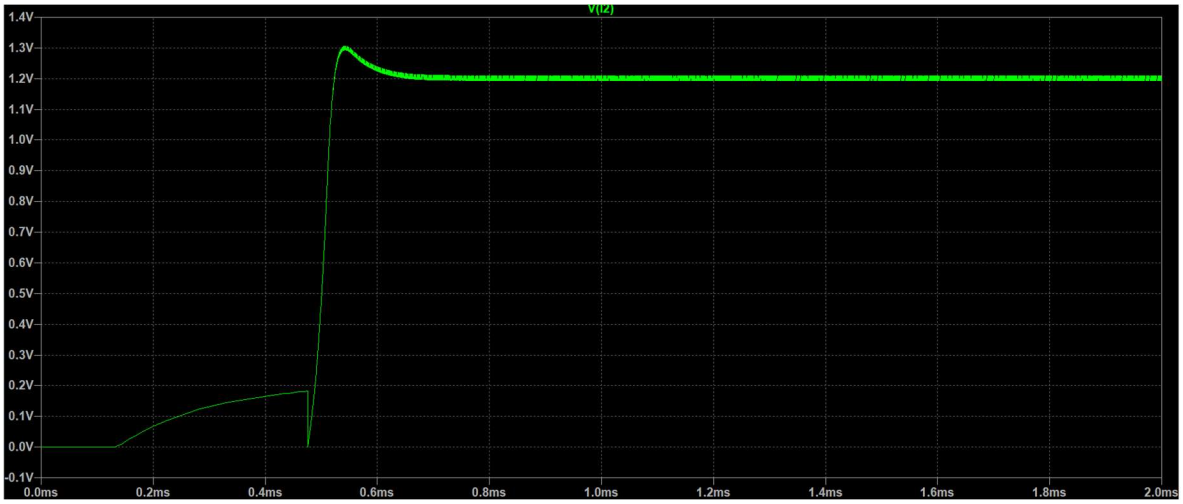


Figure 13. Buck Boost with AC to DC Source

PCB Design and Fabrication

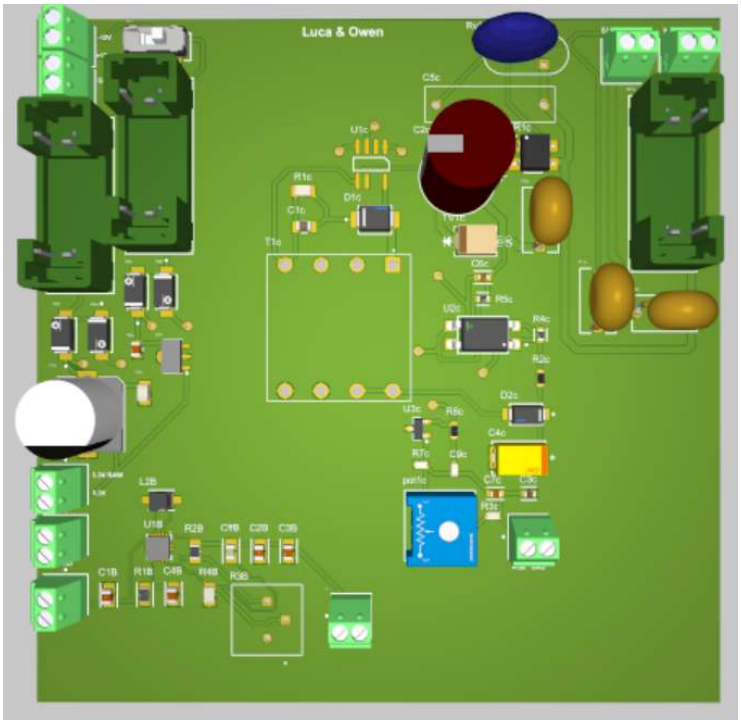


Figure 14. PCB Design

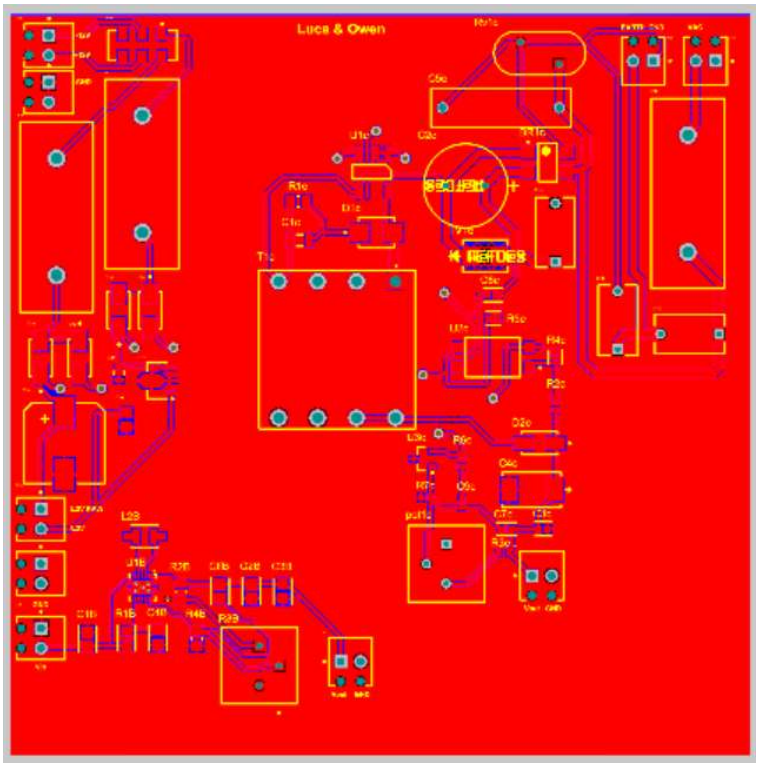


Figure 15. PCB Traces

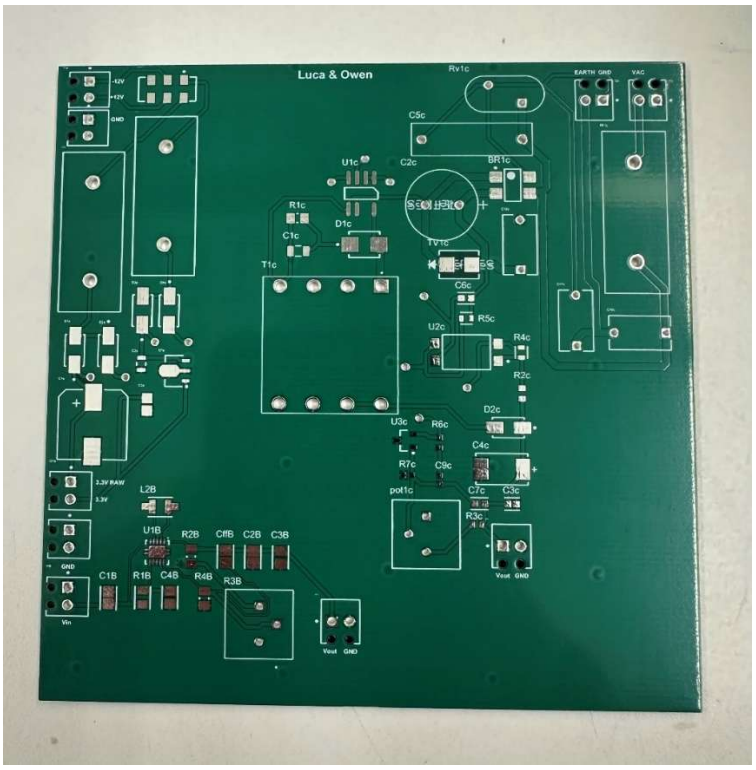


Figure 16. Fabricated PCB

Conclusion

Overall, the project successfully completed the design, simulation, and PCB fabrication of the Buck-Boost converter with an AC/DC source, as well as a flyback converter. We were able to model each circuit's schematic, as well as simulate the Buck-Boost with its AC/DC source. While final hardware testing is yet to be done, the work done throughout this project serves as a base to compare and validate our results once testing occurs. This project asked us to build a combination of circuits to work together, but more importantly tested our skills as engineers. We were able to problem-solve, found outside resources to help accomplish project requirements, and worked together to collaborate as we would in the industry.

Appendix

Bill Of Materials

Part Number / Description	Reference Designators	Quantity	Manufacturer / Source
ES1B	D1a, D2a, D3a, D4a	4	Diodes Inc.
L78L33ACUTR	U1a	1	STMicroelectronics
6.96106E+11	XF1A, XF2A, RF1C	3	Wurth Electronics
Capacitor 36V	C2a, C3a	1	Amazon
UCL1C102MNL1GS	C1a	1	Nichicon
CL-SB-22A-01T	Switch	1	Nidec Components
Molence Terminal Connector	Input/output plug	10	Amazon
Sunxeke F2AL250V Fuse	Fuse	3	Amazon
LQH32CN2R2M33L	L1B	15	Murata
TPS63000DRCRG4	U1B	1	Texas Instruments
C2012X7R1A106M125AC	C1B, C2B	2	TDK
GRM21BR61A226ME51L	C3B	1	Murata

GCM21BR91H104KA37L	C4B	1	Murata
C0805C229C1GACTU	CffB	1	KEMET
Resistor (generic)	R1B	1	DigiKey
CRCW0805174KFKEA	R2B	1	Vishay Dale
Potentiometer (generic)	R3B-pot	1	DigiKey
Resistor (generic)	R4B	1	DigiKey
HD06-T	BR1	1	Diodes Inc.
GRM188R72E152KW07D	C1	1	Murata
450BXW10MEFR10X16	C2	1	Rubycon
GRM188R61E106MA73D	C3	1	Murata
TAJE687K006RNJ	C4	1	Kyocera AVX